

YIBIN, China

WMO: 564920

Lat: 28.7688N

Lon: 104.6050E

Elev: 1654

StdP: 13.84

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.4	39.4	31.3	27.1	44.9	33.7	30.1	43.7	9.6	43.8	8.1	44.0	2.2	320	0.292

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	95.6	78.0	93.4	77.5	91.2	76.7	81.4	90.3	80.2	88.8	79.2	87.3	3.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	161.5	86.0	78.1	155.5	85.1	77.1	150.0	84.4	47.0	90.1	45.6	89.2	44.5	87.4	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.8	7.8	6.8	DB	35.1	99.0	2.3	2.3	33.4	100.7	32.0	102.0	30.7	103.3	29.0	105.0	
			WB	34.1	83.1	2.1	1.5	32.6	84.2	31.4	85.1	30.2	85.9	28.7	87.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.8	46.6	51.1	58.8	68.0	73.8	77.4	82.1	81.9	74.7	66.4	58.5	49.3	(6)
	DBStd	13.25	4.30	5.85	6.71	6.74	5.27	4.73	4.49	5.22	5.68	4.96	5.44	3.99	(7)
	HDD50	244	121	52	6	0	0	0	0	0	0	0	4	61	(8)
	HDD65	1950	569	389	214	45	3	0	0	0	1	41	202	486	(9)
	CDD50	6009	16	83	280	540	739	821	994	988	740	509	260	39	(10)
	CDD65	2243	0	0	23	135	277	371	529	523	291	86	8	0	(11)
	CDH74	19229	0	1	100	836	1910	2922	5708	5577	1928	242	5	0	(12)
	CDH80	7543	0	0	13	228	628	1002	2476	2502	655	39	0	0	(13)
(14) Wind	WSAvg	2.3	1.8	2.0	2.5	2.8	2.8	2.6	2.6	2.6	2.4	2.1	1.8	1.9	(14)
(15) Precipitation	PrecAvg	39.3	0.8	0.9	1.5	2.7	3.5	6.5	7.6	6.7	4.7	2.6	1.1	0.7	(15)
	PrecMax	57.3	1.9	1.7	3.1	5.3	7.0	15.1	15.1	12.9	10.6	4.3	3.0	1.3	(16)
	PrecMin	26.0	0.2	0.2	0.5	1.1	1.2	2.9	3.4	1.6	1.4	0.9	0.2	0.1	(17)
	PrecStd	7.2	0.4	0.4	0.7	1.0	1.5	2.9	2.8	2.8	2.5	0.9	0.6	0.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	70.5	81.5	90.2	92.9	94.4	97.6	99.3	95.5	84.6	74.6	62.1	(19)
		MCWB	51.1	56.7	63.8	69.3	70.7	75.4	78.9	78.0	77.0	70.1	63.4	53.1	(20)
	2%	DB	58.0	66.4	76.6	85.7	89.5	91.3	95.1	96.2	90.6	79.8	71.3	59.0	(21)
		MCWB	49.5	54.9	61.7	67.9	70.0	74.5	78.4	78.6	76.5	68.5	61.6	51.8	(22)
	5%	DB	54.8	62.9	72.9	82.3	86.6	88.6	93.1	93.8	86.9	76.3	68.4	56.8	(23)
		MCWB	48.5	53.7	60.3	67.0	69.8	74.2	78.4	78.3	74.8	67.2	61.0	51.0	(24)
	10%	DB	52.5	59.8	69.4	78.9	83.4	85.9	90.8	91.1	83.5	73.3	65.8	54.7	(25)
		MCWB	47.7	52.6	59.3	66.0	69.1	74.2	78.0	77.8	73.4	66.4	60.4	50.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	57.7	67.4	74.4	76.1	80.1	82.9	83.1	81.1	73.6	65.0	56.0	(27)
		MCDB	58.3	67.5	77.5	85.1	84.4	88.1	91.9	91.3	90.4	80.2	70.7	59.8	(28)
	2%	WB	50.6	56.2	64.0	70.9	74.0	78.4	81.6	81.4	78.5	70.5	63.6	53.7	(29)
		MCDB	55.4	64.0	71.6	80.3	82.3	85.8	90.5	90.3	86.3	76.1	68.5	57.2	(30)
	5%	WB	49.5	54.8	61.8	68.9	72.5	77.1	80.4	80.2	76.5	68.8	62.5	52.1	(31)
		MCDB	53.5	61.3	69.6	78.3	80.8	84.1	88.9	89.2	84.0	73.7	66.7	55.3	(32)
	10%	WB	48.3	53.2	60.0	67.2	71.1	75.7	79.5	79.2	74.7	67.3	61.2	50.9	(33)
		MCDB	51.7	58.9	67.8	76.4	79.4	82.2	87.3	87.7	81.6	71.9	64.7	53.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	8.9	11.6	13.1	13.0	11.4	12.7	12.8	10.0	8.1	7.7	6.5	(35)
		MCDBR	12.6	15.2	17.6	18.6	18.8	16.9	16.9	17.1	15.9	13.7	12.0	10.6	(36)
		MCWBR	5.8	6.0	6.0	5.6	5.4	4.9	5.2	4.7	4.6	4.3	4.3	4.8	(37)
	5% WB	MCDBR	10.6	14.0	15.7	16.2	14.2	12.7	14.5	14.9	13.7	11.2	9.9	8.2	(38)
		MCWBR	5.2	5.8	6.2	6.2	5.6	5.1	5.7	5.2	5.0	4.4	4.2	4.2	(39)
(40) Clear-Sky Solar Irradiance	taub		0.733	0.755	0.795	0.776	0.713	0.685	0.686	0.719	0.743	0.798	0.799	0.770	(40)
	taud		1.410	1.388	1.363	1.434	1.574	1.653	1.657	1.580	1.520	1.397	1.367	1.382	(41)
	Ebn at Noon		161	174	180	192	206	211	210	201	189	163	146	144	(42)
	Edh at Noon		88	96	105	100	87	80	80	85	88	94	91	87	(43)
(44) All-Sky Solar Radiation	RadAvg		515	707	967	1220	1293	1189	1500	1395	973	673	589	462	(44)
	RadStd		98	123	133	149	178	203	184	205	159	118	62	87	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) -0.59	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (5 neighbors)	+0.52	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	+200	+95

Nomenclature: See separate page